Meme Coin Analysis Dashboard

Data Engineering Project Proposal



much data.
such pipeline.

IDSC501

Team members: TBD

Doge illustration: Icons8

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01 **Ideation** — Why meme coins?

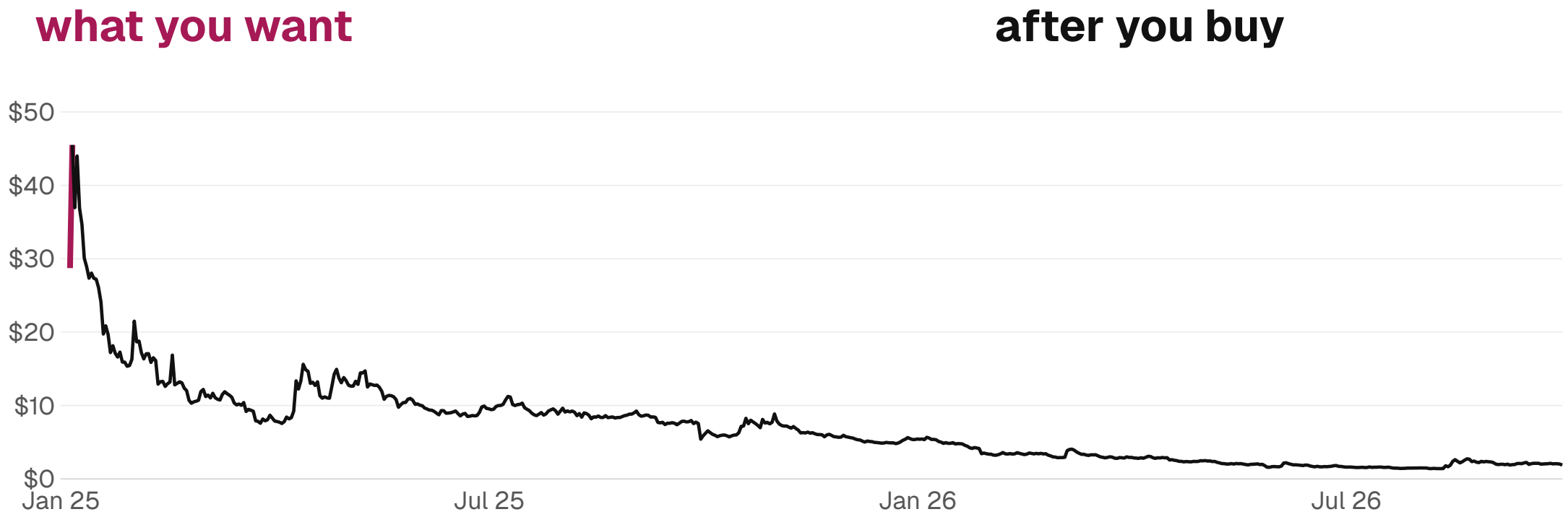
02 **Data pipeline** — Sources to dashboard

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Why meme coins?

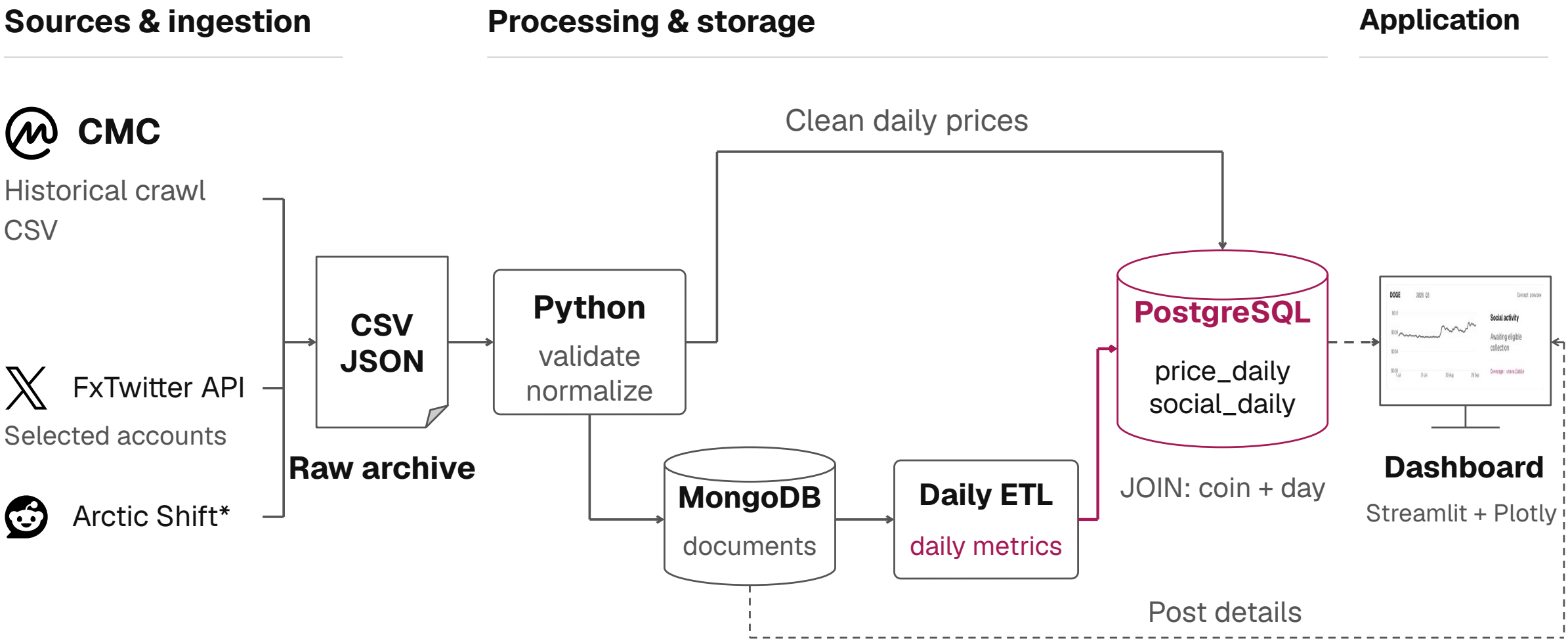
TRUMP daily close (USD)



What was the community saying?

18 Jan 2025–7 Oct 2026. Supplied daily closes. Captions are a joke, not an actual trade.

Data pipeline



Solid: batch data flow Dashed: dashboard reads * Reddit: conditional access

Data collection

14,773 price rows 10 coins · social collection planned

Source	Collection method	Output
CoinMarketCap Historical Data	**Web crawling**	Daily price CSV
X via FxTwitter	**Third-party API**	Posts + reactions in JSON
Reddit via Arctic Shift	**Archive API / files**	Posts + comments

Daily social batches, with a seven-day overlap.

Project websites: crawl links and contract details to verify sources.

Reddit awaits access conditions. Price refresh needs new CSVs or restoration of the team crawler.

Data preprocessing

From raw posts to daily features.

Raw JSON

```
{
  "id": "<post_id>",
  "text": "$DOGE…",
  "created_at": "<UTC>",
  "likes": null
}
```

Standardize

Stable post ID
UTC timestamp
Language + repost flag
Verified coin match

Daily features

Coin + UTC date
Observed post count
Mean sentiment
Coverage status

Price checks Duplicate days, OHLC rules, missing or zero values

NULL ≠ 0 Collection failure stays missing.

Schema illustration. Sentiment uses supported English text. Preserve raw inputs for replay.

Why two databases?

PostgreSQL can also store JSON.

MongoDB

Social documents

Platform-specific fields
Post and comment details
Reaction observations

PostgreSQL

Relational analytics

Keys and constraints
Prices and daily features
SQL joins and aggregation

Document model → ETL → relational model

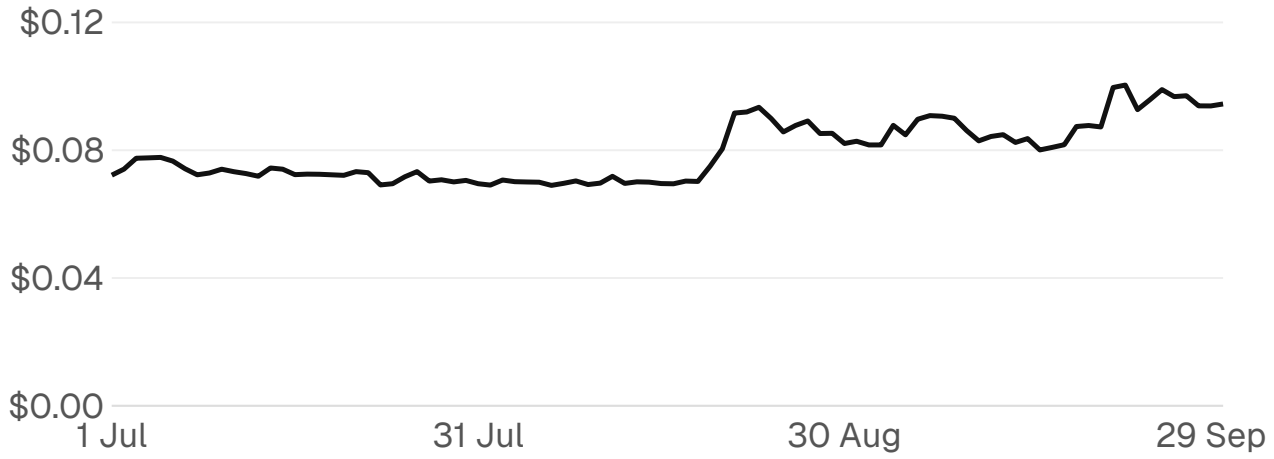
Model both SQL and NoSQL; keep social documents separate from analytical tables.

Trade-off: two databases, indexes, backups and consistency checks.

Dashboard & expected outcome

Price and social context, on one timeline.

DOGE 2026 Q3 Concept preview



Social activity

Awaiting eligible collection

Coverage: unavailable

Compare time series

Inspect posts

Check coverage

Streamlit + Plotly. Charts query PostgreSQL. Post details query MongoDB. DOGE prices use supplied data.